



Advanced Cardiovascular Technologies Lab

Department of Electrical Engineering

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Dr. Jayaraj Joseph
Associate Professor

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tinyurl.com/JayarajJoseph
<https://artsens.tech/>

To Whom It May Concern

This is to certify that Ms. Arpita Kumari, a B.Tech student in Electronics and Communication Engineering at KIET Group of Institutions, Ghaziabad has successfully completed her two-month Academic Research Internship at the Advanced Cardiovascular Technologies (ACT) Laboratory, IIT Madras, from September 10, 2025 to November 10, 2025. During her internship, Arpita worked on the Organ Project, an initiative aimed at preserving, and screening a donor's heart ex vivo. Arpita worked on developing a custom multi-channel infusion system tailored to the project's technical and operational requirements.

She designed and built the entire prototype from the ground up, which included:

- 3D CAD design of the complete system using Autodesk Fusion.
- Hardware development and integration of actuators, micro-metal gear motors with encoders, and sensor modules.
- Firmware development to ensure accurate, stable flow control and synchronization across multiple infusion channels.
- Development of a custom mobile application enabling remote operation, monitoring, and user-friendly control of the infusion system.

Through this project, Arpita gained decent hands-on experience in microcontrollers, actuators, embedded firmware, system design, and rapid prototyping. She demonstrated strong technical initiative, problem-solving ability, attention to detail, and independence in executing a complex engineering task within a short timeframe.

We appreciate her contributions to the ACT Laboratory and wish her continued success in her academic and professional pursuits.

Regards,

डॉ. जयराज जोसेफ
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